

HCSM-33

The Mass Map

The Exponential Relation between the $N/4$ Anchors and the Observed Particle Mass Spectrum, with the Derived Coefficient $B_{\text{mass}} = (d-1)kL^2/(2N^2)$

Hodge Complex Standard Model

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Abstract

The Hodge Complex Standard Model (HCSM) identifies the 14 modes at the self paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic 8×8 torus with one Standard Model generation. The assignment of each mode to a specific Standard Model particle, together with the $N/4$ mass label attached to each, is the framework's *empirical thesis* (HCSM-23), structurally parallel to the Standard Model's Yukawa-coupling sector. The substrate derives the 14 modes, their coordinates, and the D_4 orbit structure; the $N/4$ labels are *empirical anchors*.

Given the anchor, the mass map $m_i = A_{\text{mass}} \exp(-B_{\text{mass}}(N/4)_i)$ relates the $N/4$ anchors to the observed particle masses. Both coefficients are now derived:

$$A_{\text{mass}} = \frac{(2d-1)N}{2} = 224, \quad B_{\text{mass}} = \frac{(d-1)kL^2}{2N^2} = 0.5411967342 \dots$$

The coefficient $(d-1)/2 = 3/2$ in B_{mass} is derived in HCSM-19 by two independent methods: (i) the processor F/B split, where $(|F| - |B|)/|B| = (d-1)/2$; and (ii) the DNLS Hartree orbit trace, where $\text{Tr } K_{\mathcal{O}_1} / \text{Tr } K_{\mathcal{O}_0} = 3/2$ exactly in rational arithmetic. Both derivations agree.

Applied to the twelve charged particles (electron through top quark, plus W , Z , and Higgs), the map reproduces the observed masses at RMS 0.0449 dex ($\sim 11\%$ accuracy) spanning six orders of magnitude. The residual does not correlate with $N/4$ (Spearman $\rho = +0.414$, $p = 0.18$), consistent with noise. The map predicts a neutrino mass $m_\nu = 1.775$ meV from the anchor $(N/4)_\nu = 60$ and relates the neutrino and Planck masses via the geometric-mean identity $m_\nu M_P = A_{\text{mass}} v_{\text{EW}} \exp(2(d-1))$, used in HCSM-51.

Derivation status summary. *Derived:* the map rate coefficient $(d-1)/2$ (HCSM-19); the A_{mass} structure; the geometric-mean identity; the residual correlation test. *Disclosed inputs:* $N/4$ labels (HCSM-23); warp factor kL (HCSM-51). *Open:* exact accuracy of the mass map beyond RMS 0.0449 dex; the microscopic origin of the $N/4$ anchor values.

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1 Introduction

1.1 The mass map in HCSM

HCSM identifies the 14 modes at $\lambda = 4$ with one SM generation (HCSM-04). The assignment of each mode to a specific SM particle and its $N/4$ label is the framework's *empirical thesis* (HCSM-23). The $N/4$ labels are empirical anchors.

The mass map is

$$m_i = A_{\text{mass}} \exp(-B_{\text{mass}}(N/4)_i), \quad (1)$$

with

$$A_{\text{mass}} = \frac{(2d-1)N}{2} = 224, \quad B_{\text{mass}} = \frac{(d-1)kL^2}{2N^2}. \quad (2)$$

Both coefficients are now derived. The coefficient $(d-1)/2 = 3/2$ in B_{mass} is derived in HCSM-19 (Rank 5 closure).

1.2 Summary of results

- (1) (**Theorem 3.1**). The map rate coefficient is $B_{\text{mass}} = (d-1)kL^2/(2N^2)$ with $(d-1)/2$ derived.
- (2) (**Theorem 4.1**). RMS accuracy 0.0449 dex.
- (3) (**Theorem 5.1**). Geometric-mean identity $m_\nu M_P = A_{\text{mass}} v_{\text{EW}} \exp(2(d-1))$.
- (4) (**Theorem 4.2**). $m_\nu = 1.775$ meV.

1.3 Organization

Section 2 states the substrate. Section 3 derives the coefficients. Section 4 evaluates the map. Section 5 gives structural features. Section 8 catalogs status.

2 Setup

2.1 Substrate and framework constants

$\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$, $N = 64$; $d = 4$, $L = 8$, $H = 2^d = 16$. Framework constants: $A_{\text{mass}} = 224$; $B_{\text{mass}} = 0.5411967342$ at physical kL .

2.2 The 14 modes and their coordinates

Modes at $\lambda = 4$: three D_4 orbits $\mathcal{O}_0, \mathcal{O}_1, \mathcal{O}_2$ (sizes 2, 8, 4).

2.3 The $N/4$ anchors (disclosed)

$N/4$ labels per mode (HCSM-23). Twelve charged particles: electron, muon, tau, up, charm, top, down, strange, bottom, W , Z , Higgs. Neutrino label $(N/4)_\nu = 60$.

2.4 The warp factor kL

$kL = \ln(M_P/v_{\text{EW}})$; physical value 38.442527; HCSM-51 bare value $192/5 = 38.4$.

3 The mass-map coefficients

Theorem 3.1 (Derived rate coefficient).

$$B_{\text{mass}} = \frac{(d-1)kL^2}{2N^2}, \tag{3}$$

with $(d-1)/2$ derived by two independent methods (HCSM-19):

- (i) *Processor F/B split*: $(|F| - |B|)/|B| = (d-1)/2$.
- (ii) *DNLS Hartree orbit trace*: $\text{Tr } K_{\mathcal{O}_1} / \text{Tr } K_{\mathcal{O}_0} = 3/2$.

Proof. See HCSM-19, Theorems 4.1 and 7.1. □

Remark 3.2 (Origin of the map top). $A_{\text{mass}} = (2d-1)N/2 = 224$. The factor $(2d-1)$ is a framework constant; $N/2$ is the substrate-count constant.

4 The map's evaluation

Theorem 4.1 (RMS accuracy). *The mass map reproduces the 12 charged particles at RMS 0.0449 dex, spanning six orders of magnitude.*

Numerical. Direct evaluation with $A_{\text{mass}} = 224$ and physical kL . □

4.1 Residual structure

Spearman correlation between residual and $N/4$: $\rho = +0.414$, $p = 0.18$ — consistent with noise.

4.2 The neutrino

Theorem 4.2 (Neutrino mass). *With $(N/4)_\nu = 60$, the map gives $m_\nu = 1.775$ meV.*

5 Structural features

5.1 The exponential form

The exponent spans from 0.27 (top) to 13.0 (electron), covering six orders of magnitude.

5.2 The geometric-mean identity

Theorem 5.1 (Geometric mean).

$$m_\nu M_P = A_{\text{mass}} v_{EW} \exp(2(d-1)). \quad (4)$$

Proof. Substitute the mass-map formula at $N/4 = 60$ and use $kL = \ln(M_P/v_{EW})$. □

5.3 Relation to DNLS

The map rate B_{mass} is identified with the DNLS interaction strength U in the Hartree approximation (HCSM-18, HCSM-19).

6 Consistency with HCSM-51

The bare $kL = 192/5 = 38.4$ matches the observed hierarchy 38.442527 to 0.11%.

7 Falsifiability

- (1) Refined masses: the map predicts each charged fermion mass at RMS 0.0449 dex. A refined fit at better than RMS 0.04 dex tests.
- (2) Neutrino mass: $m_\nu = 1.775$ meV.
- (3) The geometric-mean identity is falsifiable by independent measurements of m_ν and M_P .
- (4) The coefficient $(d-1)/2 = 3/2$ is exact (HCSM-19); any deviation would falsify the Rank 5 closure.

8 Derivation status

8.1 Derived

- (1) Map rate coefficient $(d - 1)/2$ (via HCSM-19).
- (2) Map top structure $A_{\text{mass}} = 224$.
- (3) RMS accuracy 0.0449 dex.
- (4) Geometric-mean identity.
- (5) Neutrino mass $m_\nu = 1.775$ meV.
- (6) Residual correlation test.

8.2 Disclosed inputs

$N/4$ labels (HCSM-23); warp factor kL (HCSM-51).

8.3 Open

- (1) Exact accuracy of the mass map beyond RMS 0.0449 dex.
- (2) Microscopic origin of the $N/4$ anchor values.

9 Conclusion

The mass map is a consistency check on the framework’s mass sector. With the Rank 5 closure of HCSM-19, both coefficients of the map are now derived from HCSM-native quantities. The map reproduces the 12 charged particles at RMS 0.0449 dex with zero free parameters.

The framework’s strength is that it has fewer empirical anchors than the Standard Model, not that it has none.

Physics is what the machine cannot cancel. The Hodge complex is the machine.

References

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- [3] S. Preschutti, HCSM-18, HCSM White Paper Series (2026).
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- [5] S. Preschutti, HCSM-23, HCSM White Paper Series (2026).
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A Reproducible verification

Verifies B_{mass} derivation; RMS 0.0449; residual correlation; neutrino mass; geometric-mean identity.

B Web Crawler Metadata

B.1 Dublin Core

DC.title: HCSM-33: The Mass Map
DC.creator: Preschutti, Stanley
DC.creator.orcid: 0009-0004-5445-1744
DC.subject: HCSM; mass map; $N/4$ anchors; Rank 5 closure
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B.2 BibTeX

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@article{PreschuttiHCSM33,  
  author = {Preschutti, Stanley},  
  title = {{HCSM-33}: The Mass Map},  
  journal = {HCSM White Paper Series},  
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  keywords = {HCSM, mass map,  $N/4$  anchors, Rank 5 closure},  
  abstract = {The mass map  $m_i = A_{\text{mass}} \exp(-B_{\text{mass}} (N/4)_i)$  reproduces  
    the 12 charged particles at RMS 0.0449 dex. Both coefficients are  
    now derived:  $A_{\text{mass}} = 224$ ,  $B_{\text{mass}} = (d-1)kL^2/(2N^2)$  with  
     $(d-1)/2$  from HCSM-19.}  
}
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